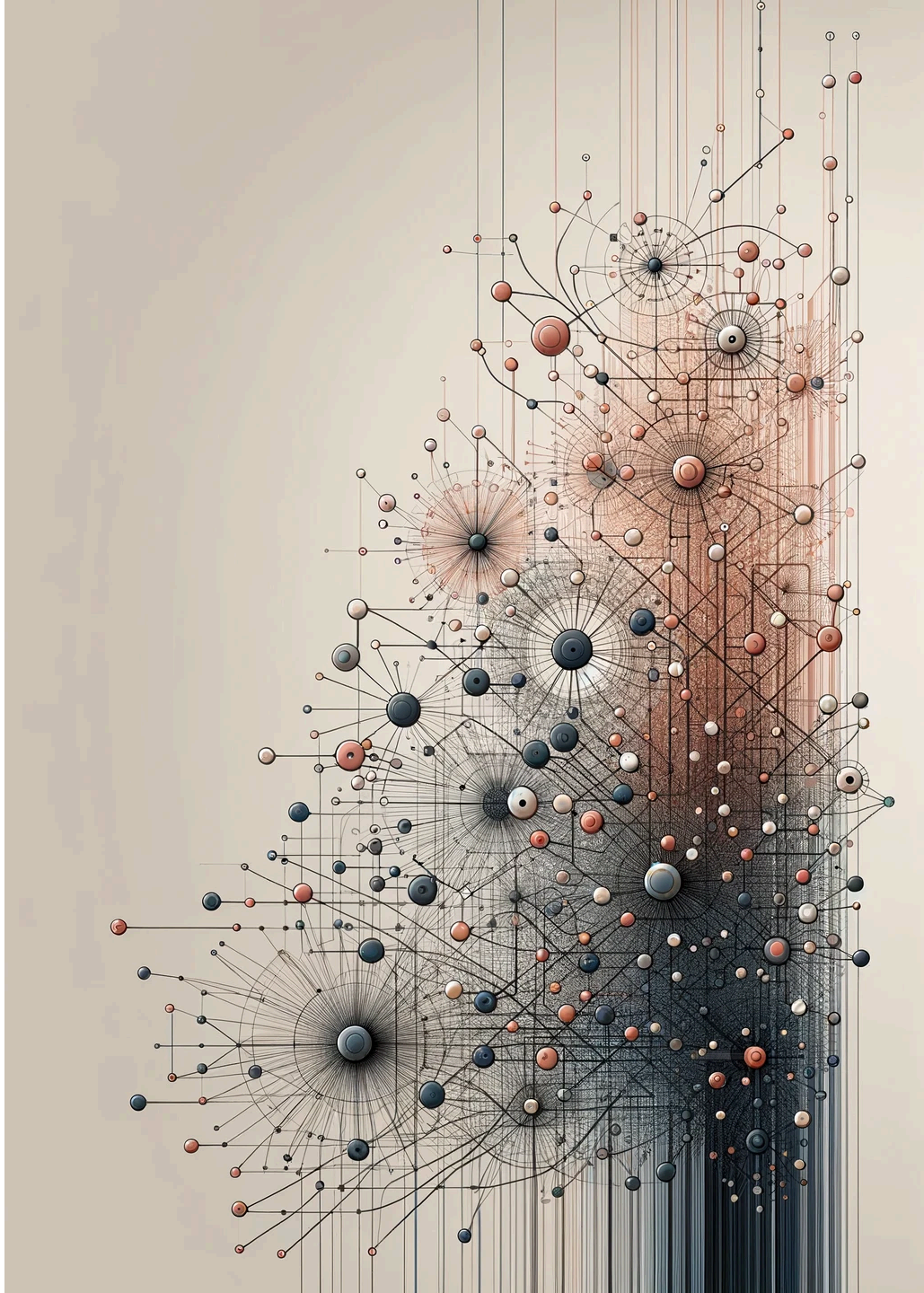
UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS10 - Preferential Attachment Extensions

Prof. Rossano Schifanella



Outline

- Why BA is not enough
- Attractiveness and baseline visibility
- Fitness and node quality
- Random walk and local closure
- Copying and imitation mechanisms
- Which extension to use

What needs to be fixed after BA

The BA model explains hub emergence, but it leaves several empirical problems open:

- zero-degree or source-node disadvantage
- strong age dominance
- low clustering
- unrealistic reliance on global degree information
- fixed exponent and limited mechanism flexibility

This lecture studies extensions that address these failures one by one.

A useful strategy for reading extensions

For each model, ask the same questions:

- what empirical failure of BA motivates it?
- what rule is changed?
- what structural property changes as a result?
- what does the extension still fail to explain?

This keeps the comparison mechanistic rather than descriptive.

From BA failure to model extension

BA limitation	Extension
zero-degree / source-node disadvantage	**Attractiveness**
oldest nodes dominate too strongly	**Fitness**
low clustering and unrealistic global information	**Random Walk**
imitation and duplication are central	**Copy model**

The order of the lecture follows this map: baseline visibility, then quality, then local search, then imitation.

Baseline visibility: the attractiveness model

The problem: zero degree means zero chance

Pure preferential attachment creates a severe problem in some settings.

If attachment probability is proportional only to degree, then a node with degree zero has no chance of receiving future links.

This is especially problematic in directed settings, where new nodes often start with in-degree zero.

The attractiveness model

The rule becomes:

$$P(i \leftrightarrow j) = \frac{A + k_j}{\sum_l (A + k_l)}$$

where $A > 0$ is a baseline attractiveness.

Every node now has a non-zero probability of being selected, even before it has accumulated degree.

What the attractiveness parameter changes

The parameter A acts as a degree offset.

- if $A = 0$, we recover BA
- if A is large, degree differences matter less
- intermediate values preserve reinforcement but reduce first-mover dominance

This turns a rigid model into a tunable one.

Structural consequence of attractiveness

The model still produces a power-law degree distribution, but the exponent becomes adjustable:

$$\gamma = 3 + \frac{A}{m}$$

So the tail can decay faster than in the standard BA model.

This is useful when empirical data are broad, but not as broad as BA predicts.

What the attractiveness model solves, and what it does not

It addresses:

- zero-degree exclusion
- overly strong first-mover advantage
- fixed exponent $\gamma = 3$

It does not address:

- low clustering
- local search or triadic closure
- node-specific quality differences

Quality differences: the fitness model

The problem: older nodes are not always the winners

In many real systems, young nodes can overtake old ones.

Examples:

- a new web platform can overtake older sites
- a recent paper can become more influential than many classic ones
- a new product can dominate despite arriving late

Degree alone is not enough. Node quality also matters.

The fitness model

Assign each node j an intrinsic fitness η_j.

Then the attachment rule becomes:

$$P(i \leftrightarrow j) = \frac{\eta_j k_j}{\sum_l \eta_l k_l}$$

Connectivity still matters, but it is modulated by node quality.

What fitness changes conceptually

The model combines two forces:

- **cumulative advantage** through degree
- **intrinsic advantage** through fitness

This allows late but high-quality nodes to become hubs.

The competition is no longer driven only by age.

Structural consequence of fitness

The degree distribution is no longer governed by a single universal exponent.

Instead, outcomes depend on the distribution of fitness values:

- bounded or moderate fitness gives more balanced competition
- highly unequal fitness can produce extreme concentration
- in some regimes, one node can capture a finite fraction of all links

What the fitness model solves, and what it does not

It addresses:

- age dominance
- unrealistic “oldest nodes must win” behavior
- quality-sensitive growth

It does not address:

- low clustering
- local search and triangle formation
- the cost of assuming node-specific latent quality

Local information: the random-walk model

The problem: BA uses global degree information

The BA model implicitly assumes that new nodes know how connected every old node is.

That is often unrealistic.

In many systems, new links are formed through **local exploration**, not global ranking.

A second problem: BA creates too few triangles

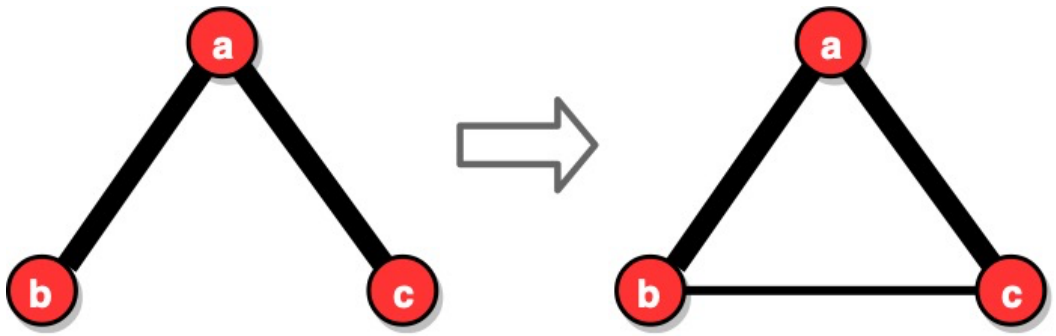
Real social and information networks often have strong local closure.

But BA attaches to important nodes without asking whether those targets are neighbors of one another.

So it creates hubs, but not enough triadic closure.

The random-walk model

1. choose a random target node j
2. create the first edge to j
3. for later edges:
   - with probability p, choose a neighbor of j
   - with probability $1 - p$, choose a random node



This adds a local-search mechanism and encourages triangle formation.

Why the random-walk rule works

A random walk reaches high-degree nodes more often because they have more incident edges.

So preferential attachment appears **implicitly**, without explicitly computing degree.

At the same time, walking through neighbors creates local closure and raises clustering.

What the random-walk model solves, and what it does not

It addresses:

- unrealistic global-information assumptions
- low clustering
- lack of triadic closure in BA

It does not fully address:

- node quality differences
- all forms of empirical degree variation
- other mechanisms such as aging or deletion

Imitation and duplication: the copy model

The problem: many new links are copied, not discovered from scratch

In several systems, new nodes imitate existing connection patterns.

Examples:

- new web pages copy hyperlink structures from related pages
- duplicated genes inherit interaction patterns
- new content often reuses reference structures from existing content

This suggests an imitation-based growth mechanism.

The copy model

At each step:

1. add a new node
2. choose an existing target node j uniformly at random
3. with probability p, connect directly to j
4. with probability $1 - p$, connect to a neighbor of j

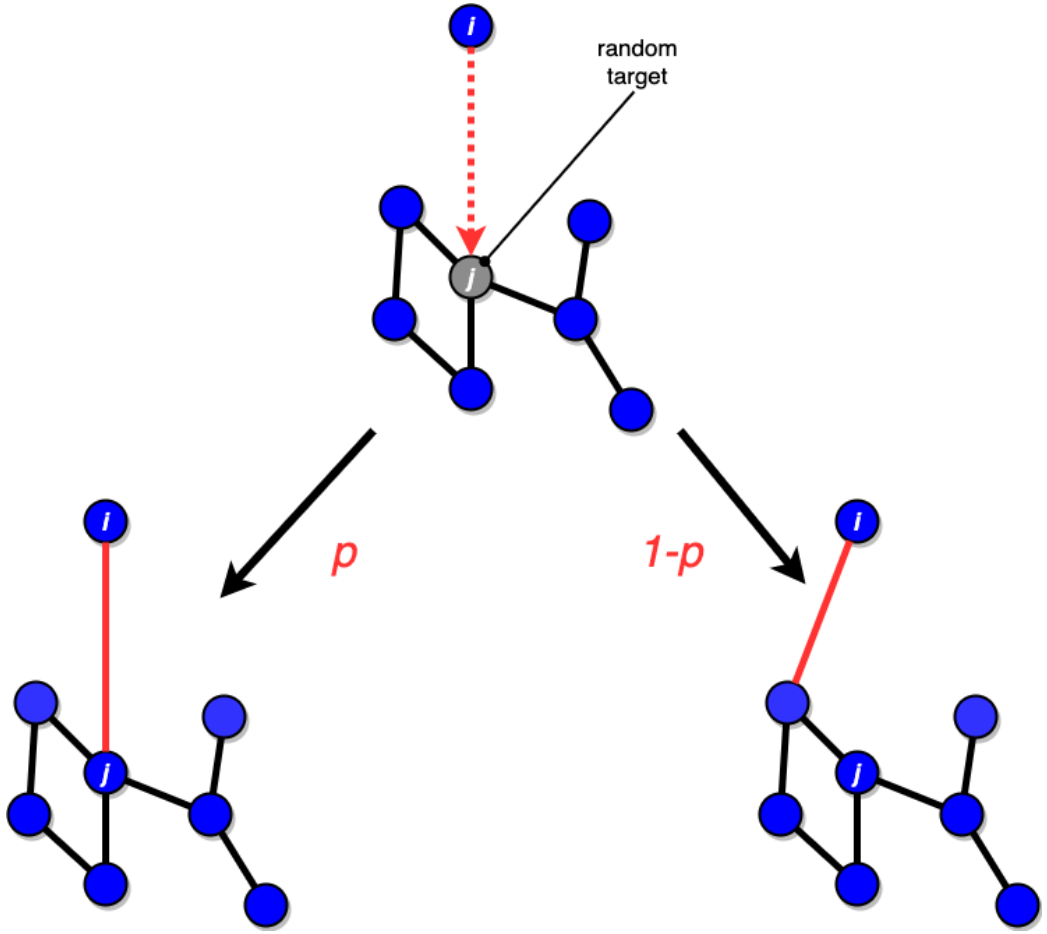
Unlike the random-walk model, the new node connects to **either** the target or its neighbor, not both.

Copying is not the same as triadic closure

The copy model generates implicit preferential attachment through imitation.

But it does not explicitly force triangle closure in the way the random-walk rule does.

This difference matters for clustering.



Structural consequence of the copy model

The copy model can generate a tunable power-law tail:

$$\gamma = \frac{2-p}{1-p}$$

So it preserves broad degree heterogeneity while capturing imitation-based growth.

Its clustering is typically higher than BA, but lower than explicit triadic-closure models.

What the copy model solves, and what it does not

It addresses:

- imitation-based growth
- local rather than global information use
- tunable degree heterogeneity

It does not fully address:

- strong triangle closure in the social-network sense
- node-quality differences
- all sources of mesoscale organization

Comparing the extensions

Model	BA failure it addresses	Key mechanism	Degree effect	Clustering effect	Best use case
Attractiveness	zero-degree exclusion, fixed exponent	baseline offset $A + k$	tunable $\gamma > 3$	still low	visibility / initial appeal
Fitness	age dominance	node quality η_j	quality-dependent tail	still low	meritocratic competition
Random walk	global information, low	local search + neighbor step	implicit PA	high	social closure / local search

Which model should we use?

Use the model that matches the missing mechanism in the empirical system.

- if degree heterogeneity is the main goal: **BA** or **Attractiveness**
- if node quality must matter: **Fitness**
- if local closure is essential: **Random Walk**
- if imitation or duplication is central: **Copy model**

The right model is chosen by mechanism, not by visual resemblance alone.

A general lesson from these extensions

Preferential attachment is not one single model.

It is a family of growth mechanisms in which current success influences future attachment.

Different extensions change:

- what information new nodes use
- whether node attributes matter
- whether local closure is present
- how broad the resulting degree hierarchy becomes

Next step: interpreting rich-get-richer claims

Once several reinforcement mechanisms are on the table, a more careful question becomes necessary.

When we observe a broad distribution or a rich-get-richer pattern:

- is it really a power law?
- which mechanism is actually responsible?
- what should and should not be inferred from the data?

This leads to the next conceptual step: how to interpret rich-get-richer and power-law claims more carefully.

Reading material

References

[ns1] **Chapter 5 (Network Models)**

Q&A